Setting Up A Network: Hardware

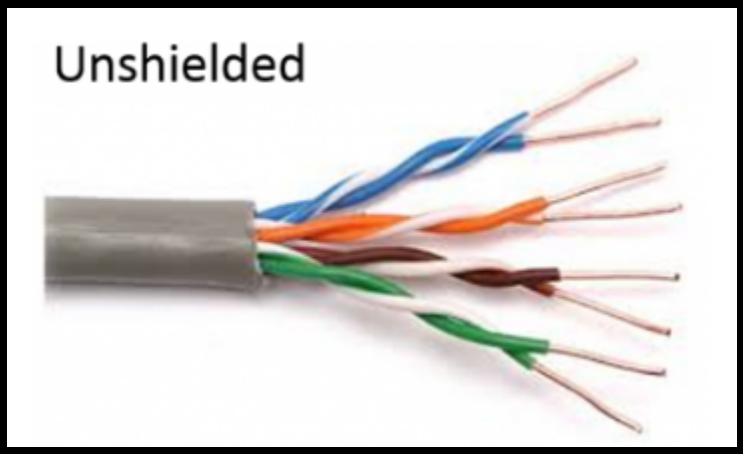
Setting up a small network, called a LAN, requires a few pieces of hardware. You'll need the node devices (a computer), a router and ethernet cables.

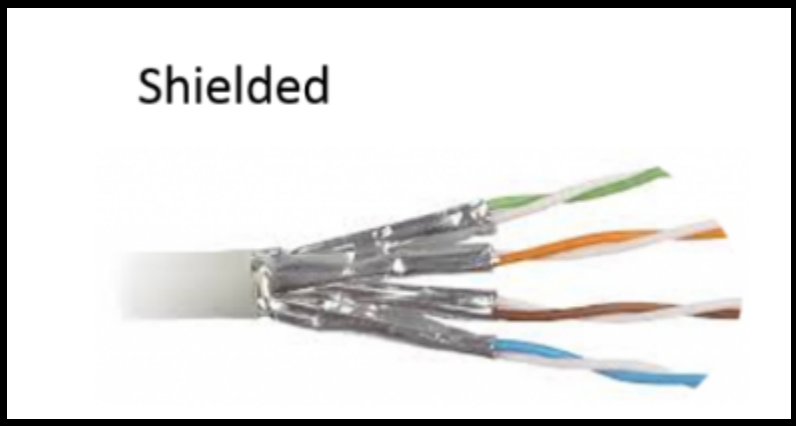
You've been supplied with computers. You will be supplied with a router. You will build your cables.

Building Ethernet Cables

In order to build ethernet cables you will require the following:

UTP cable→_Unshielded Twisted Pair cable





This cable comes with 4 pairs of twisted copper wires. The paired wires are coloured, both solid and alternating with white.The colours are as follows:

Orange-orange white
Green-green white
Brown-brown white
Blue-blue white

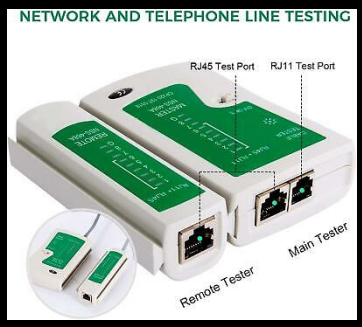
Wire cutter and crimper



RJ45 connectors



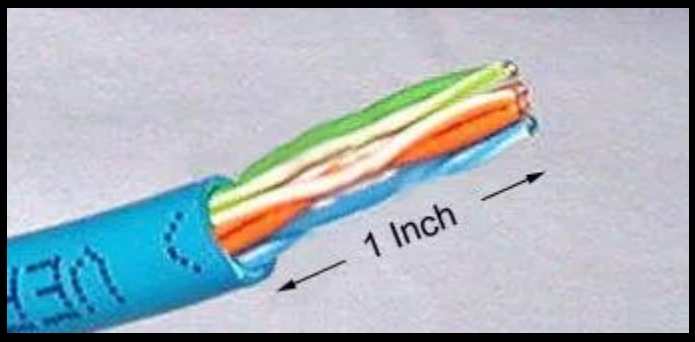
Ethernet cable testing tool



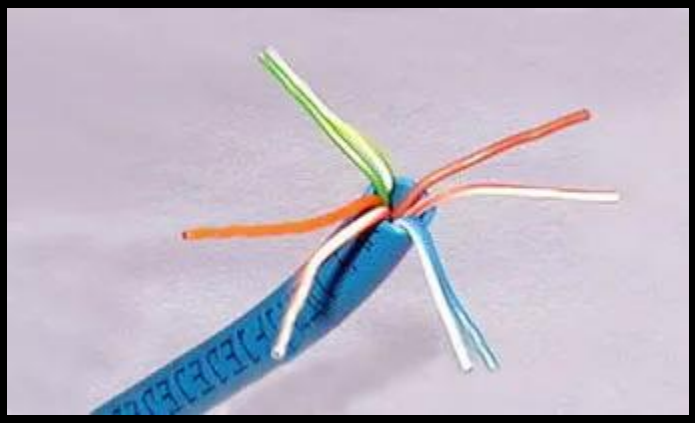
To build an ethernet cable you will first need to measure how long the cable needs to be. Then using the cutting portion of your wire cutting tool cut the cable to length. Using the same tool cut off about 1 inch off the ends of the sheathing of the cable.



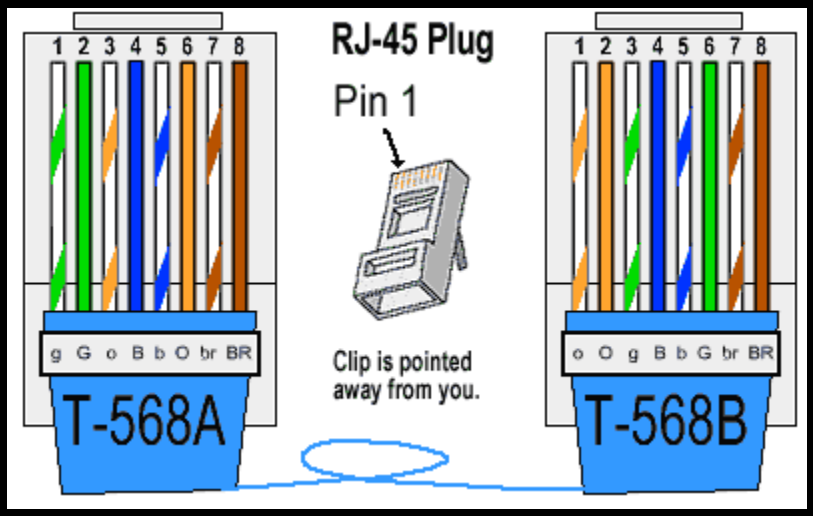
This will expose the 4 pairs of twisted wires.



Untwist the wires from each other and straighten them out.

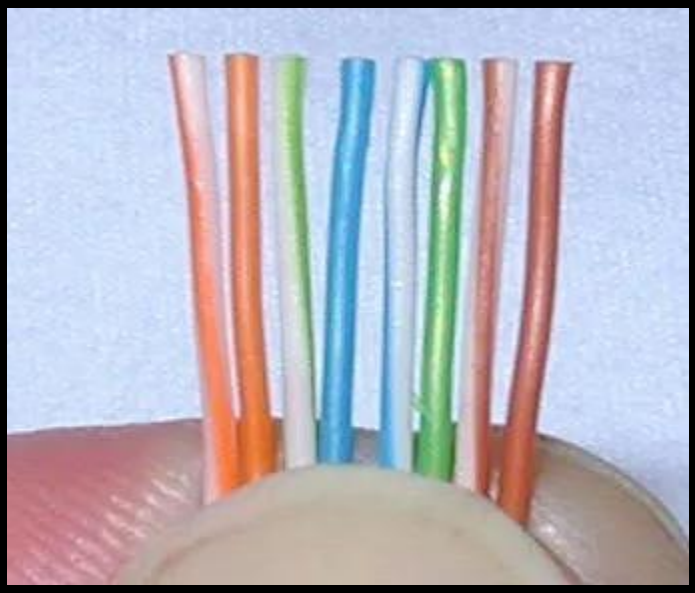


You now must decide which pinout format you will use (T568A or T568B):



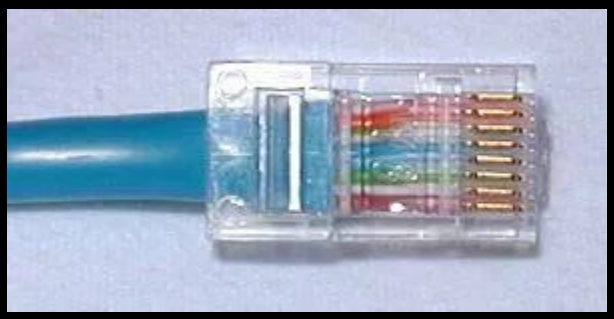
To make a straight through or patch cable i.e. one that connects two different devices i.e. your computer and router, you simply choose one format and apply it to both ends of the cable. To make a crossover cable i.e. one that connects two similar devices you use T568A on one end and T568B on the other.

Straighten out the wires in the format you have selected and then snip the ends so their straight across (the pic shows T568B which is the most common format used):



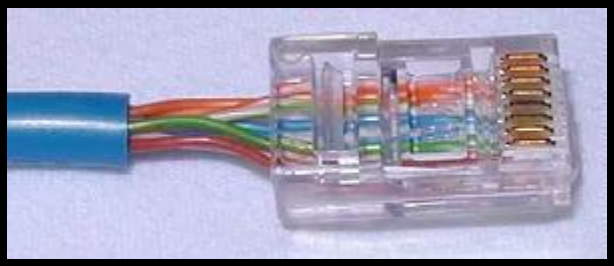
Slowly push the connector onto the wires, making sure each wire slides through its own channel to the end of the connector, just below the gold contact pins.





Note the orientation of the connector relative to the wires. Note that by looking at the connector from the top and sides you can see whether or not the wires have reached their ends.

Some common errors:



Too much sheathing is removed.



Wires too short and not able to reach the ends.

Once everything looks good slide the connector into a crimping tool:



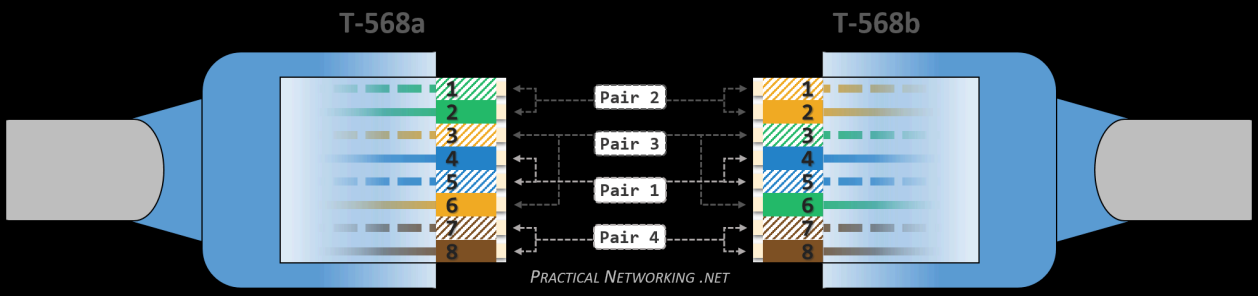


Make sure the connector is all the easy in. Press down hard on the crimper and the gold contact pins should pierce the coloured wires.

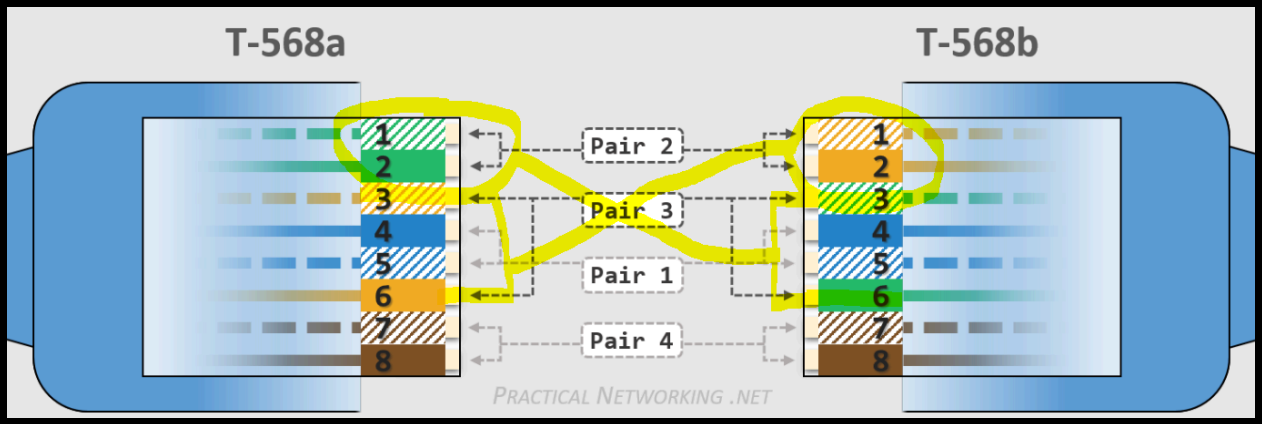
Repeat all of this at the other end.

Test the cable using the ethernet testing tool. Most cables/networks will operate as long as wires 1,2 and 3,6 are connected properly (green and orange cables, the blue and brown set aren't used).





Important Note: normally only the green and orange pins need to be successfully connected in order for an ethernet cable to work (unless you need a Gigabit+ connection). For example, if your multimeter tester flashes a green signal for pairs 1+2, 3+6. 7+8 and not 4+6 the cable should still work. Why? Because the green (1+2) and orange (3+6) pairs are connected properly. Even though the brown pairs (7+8) are connected it doesn't matter and the 4+5 blue pair doesn't matter at all.



The following shows a ethernet testing tool with a bad ethernet cable:



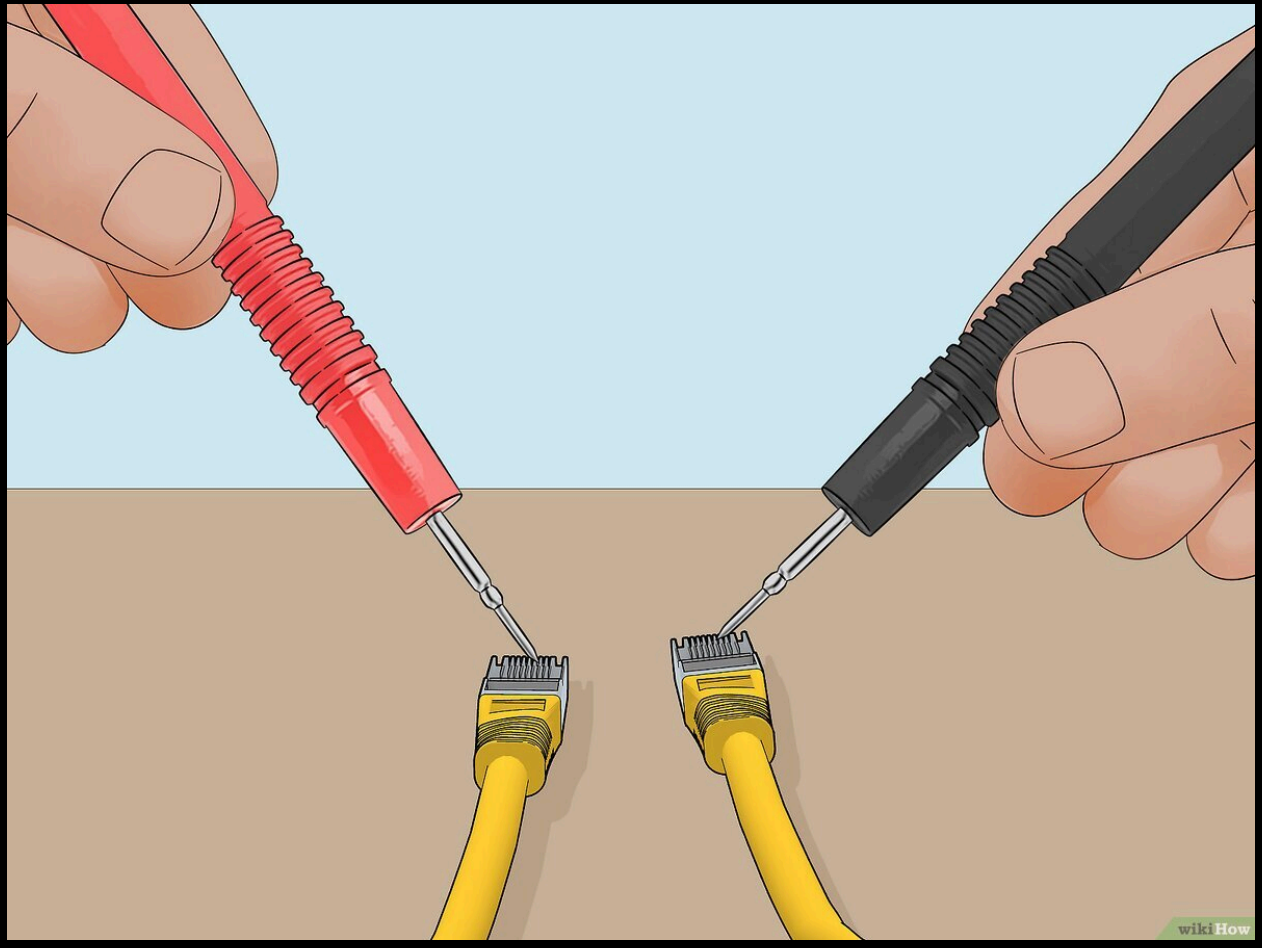
Test Method #2: Use a _**multimeter**

We can use a multimeter to test the _**_continuity** between each set of pins. Continuity testing simply refers to testing if two ends of the same wire are connected and will allow the flow of electrons from one end to another.

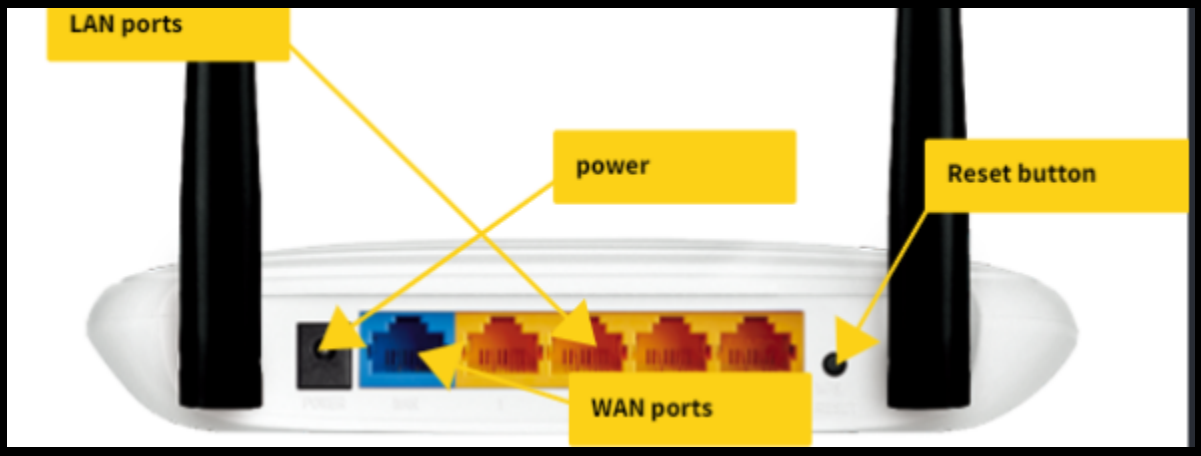
To test continuity with a multimeter first start by making sure the red and black probes are connected to the correct ports on the multimeter (see image). Also make sure to dial the multimeter's setting selector to the continuity setting (see image).



Touch the two probes together and you should __ hear a beep. This means continuity testing will work. Finally touch each probe to each end of the individual (same coloured) wires on both ends of the ethernet cable. If you get a sound with all 8 wire pairs your ethernet cable should work. If you only get a response with the orange and green pairs, the cable should work. Watch this video as a reference: https://youtu.be/84yhs3FilKc?si=6E3mBnEFwbnu4NsR



Once the cables have been constructed plug one end into a LAN port on your router(**TL-WR841N**) and the other end into the back of your computer. Reset the router by pressing in the reset button for 5-10 seconds (check the documentation for more specific information).



Turn your computer and the router on.

Look at the little LED light on the LAN port. If it is blinking then you have probably connected your computer correctly.



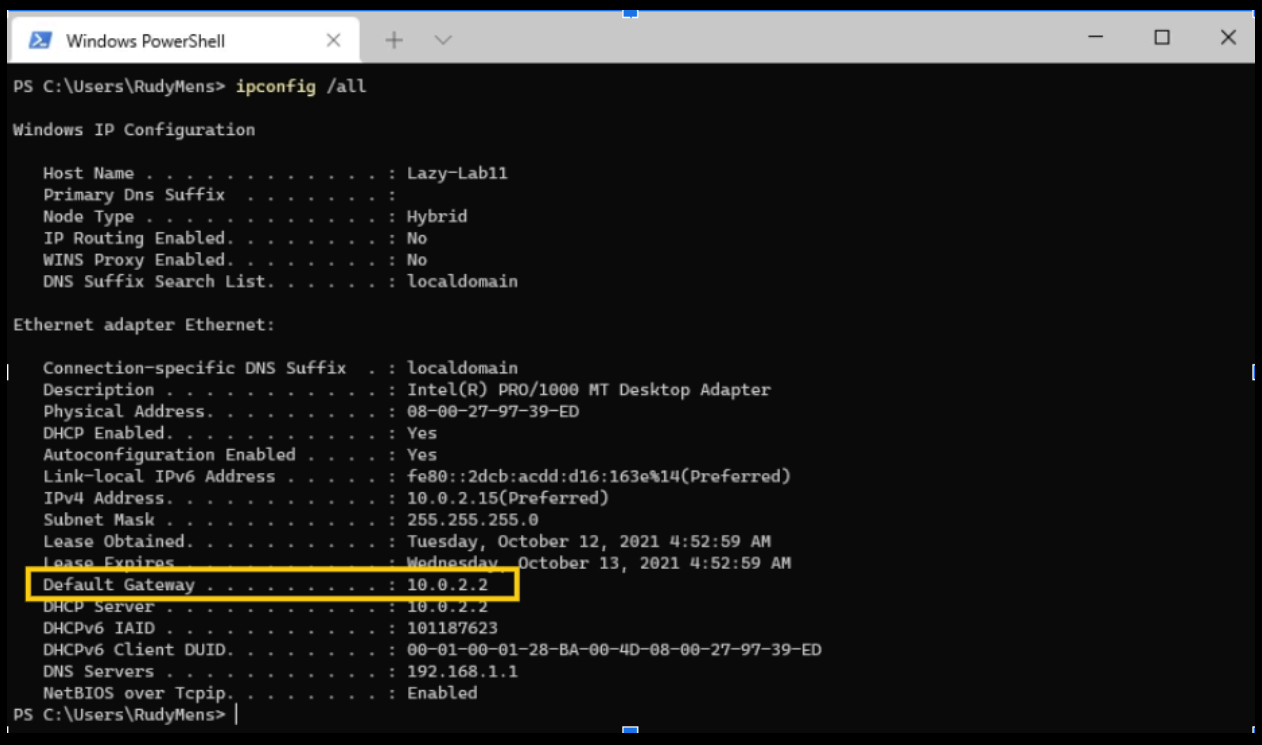
Windows Command Prompt
This is a program that gives you command line access to your computer. Being able to interact with it is important.

There are many ways to access the command prompt. Use this one:
Press **Win + R** to open the Run box, then type "cmd" and hit **Enter** to open it.

Once opened you can now use a whole host of text based commands to control your computer. Use the following command

Ipconfig /all

Once typed into the command prompt, press Enter. You will see something as follows:



Look for Default Gateway.

This is the IP address (we will learn more about IP addresses soon) of your router (in the above example this would be 10.0.0.2).

Go to your browser and type this IP address into the address bar and then hit enter.You should see your router's login page show up.